

# Collision Geometry of Joint Spectra

*Affine Branch Arrangements, Exact Finite Censuses, and Conditional Spectral Quotients*

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## Abstract

**Problem.** A finite joint spectrum can be observed at two resolutions: a resolved set of joint spectral points and a coarser spectrum obtained by a linear functional. We ask how collisions of affine branches organize the resulting quotient, explicitly separating general finite-point theorems from a numerically registered operator realization.

**Approach.** For a fixed finite arrangement  $P = \{(q_i, h_i)\}_{i=1}^N \subset \mathbb{R}^2$ , we study  $L_\alpha(q, h) = \alpha q + (1 - \alpha)h$  and the equivalence relation induced by equal  $L_\alpha$  values. We prove the collision formula, finiteness of the critical set, constancy of branch order on its complementary chambers, and the spectral quotient theorem for commuting Hermitian realizations. Rational coordinates give rational finite collision parameters. A two-parameter rectangular family provides an exact control beyond the main example. We then perform a complete exact census for one explicitly declared nine-point rational arrangement.

**Results.** For the explicit arrangement  $P_9$ , the 36 unordered pairs split into 2 parallel, 10 interior, 15 endpoint, and 9 exterior pairs. On  $[0, 1]$  the interior critical parameters are  $2/7$ ,  $2/5$ ,  $1/2$ ,  $2/3$ , and  $4/5$ . The layer-count drops are respectively 1, 2, 2, 3, and 1, so  $\alpha = 2/3$  is the unique interior parameter of maximal drop. Its nontrivial quotient classes contain S5, S6, and S7, and S8 and S9, respectively. For the rectangular family with side parameters  $a, b > 0$ , the sole interior collision occurs at  $\alpha = b/(a + b)$  and produces one two-point class.

**Boundary.** In the declared complex128 Rubik computation, coordinate-matched labels, ranks, and projectors are stable across the tested clustering tolerances, and the maximum raw-to- $P_9$  table discrepancy is below  $10^{-15}$ . Machine-zero commutator and projector residuals support this registration but do not prove exact commutation or exact rational joint eigenvalues. The six exact averaging layers are the  $L_{2/3}$  collision quotient of  $P_9$  only when exact operator admissibility, exact labelled joint registration, and the exact averaging identity all hold. No parameter-dependent or maximality conclusion is drawn.

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## Notation and Claim Layers

| Symbol                           | Meaning                                                   |
|----------------------------------|-----------------------------------------------------------|
| $P = \{p_i\}_{i=1}^N$            | fixed finite set of distinct points $p_i = (q_i, h_i)$    |
| $L_\alpha(q, h)$                 | $\alpha q + (1 - \alpha)h$                                |
| $\lambda_i(\alpha)$              | affine branch $L_\alpha(p_i)$                             |
| $\mathcal{C}(P)$                 | finite collision set in $\mathbb{R}$                      |
| $i \sim_\alpha j$                | equality $L_\alpha(p_i) = L_\alpha(p_j)$                  |
| $P/\sim_\alpha$                  | collision quotient at $\alpha$                            |
| $N_P(\alpha)$                    | number of quotient classes                                |
| $\Delta_P(\alpha)$               | layer-count drop $ P  - N_P(\alpha)$                      |
| $m_{\text{pair}}(\alpha)$        | number of colliding unordered pairs at $\alpha$           |
| $P_9$                            | explicit weighted nine-point rational arrangement         |
| $Q, H$                           | commuting Hermitian operators in the exact general theory |
| $Q_{\text{num}}, H_{\text{num}}$ | numerically constructed Rubik QT/HT averages              |
| $A_{18}$                         | canonical 18-generator Rubik average                      |

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The paper uses four claim-status levels:

1. **Theorem.** The statement uses exact objects, displayed hypotheses, and a proof. A conditional corollary is available only after every displayed hypothesis is established.
2. **Computational Certificate.** The statement is a reproducible finite computation tied to declared operators, tolerance policy, and a result record. Residual agreement does not discharge an exact theorem hypothesis.
3. **Computational Observation.** The statement records a bounded finite comparison under a declared threshold. The observed pattern supplies no general implication or promotion theorem.
4. **Research Program.** The statement is an explicitly open construction or question. It is not used as an assumption, lemma, or explanation of an established result.

The classification is fail-closed. Exact arithmetic performed after a numerical registration does not promote the registration, and a conditional corollary does not certify its own assumptions. In particular, the exact finite census of  $P_9$  and the conditional Rubik corollary lie in the Theorem level for different reasons: the former is proved directly from displayed rational data, whereas the latter applies only under the exact assumptions listed in Section 7.

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## 1 Introduction

Let a fixed finite joint-spectrum arrangement be given by

$$P = \{(q_i, h_i)\}_{i=1}^N \subset \mathbb{R}^2.$$

The one-parameter family of linear observables

$$L_\alpha(q, h) = \alpha q + (1 - \alpha)h$$

assigns to each point an affine branch

$$\lambda_i(\alpha) = L_\alpha(q_i, h_i).$$

At a generic parameter the branch values are distinct. At a critical parameter, two or more values coincide and the resolved point set is replaced by a quotient partition. If  $P$  is the exact joint spectrum of commuting Hermitian operators  $Q$  and  $H$ , this quotient is precisely the eigenspace decomposition of  $A(\alpha) = \alpha Q + (1 - \alpha)H$ .

This observation is elementary, but three logically distinct tasks must not be conflated:

- proving the general finite-point quotient theorem;
- classifying one explicitly declared rational arrangement exactly;
- registering a numerical operator computation against that arrangement under declared tolerances.

The present paper separates these tasks. Its contributions are:

1. a self-contained collision-quotient theorem for fixed finite arrangements;
2. a rationality proposition for finite collision parameters;

3. an exact rectangular control family and a complete phase census on  $[0, 1]$  for a weighted nine-point rational arrangement  $P_9$ ;
4. a claim-status-aware computational certificate for the Rubik QT/HT clusters;
5. a conditional corollary identifying the Rubik averaging layers with the exact quotient under three separately stated exact assumptions.

The general theory does not require Rubik data. Conversely, the numerical registration does not become exact merely because the downstream finite-point arithmetic uses rational numbers. This direction of dependence is the central claim discipline of the paper.

The fixed-arrangement output may be applied pointwise once an exact or explicitly registered joint spectrum is available. By contrast, any parameter-dependent application requires a certified normal spectral chart and coherent projector continuation.

## 2 Finite Collision Arrangements

**Definition 2.1** (Affine branch arrangement).

Let  $P = \{p_i = (q_i, h_i)\}_{i=1}^N \subset \mathbb{R}^2$  be finite with distinct points. For  $\alpha \in \mathbb{R}$ , define

$$L_\alpha(q, h) = \alpha q + (1 - \alpha)h, \quad \lambda_i(\alpha) = L_\alpha(p_i).$$

Thus

$$\lambda_i(\alpha) = h_i + \alpha(q_i - h_i).$$

**Definition 2.2** (Collision set and quotient).

The collision set is

$$\mathcal{C}(P) = \{\alpha \in \mathbb{R} : \lambda_i(\alpha) = \lambda_j(\alpha) \text{ for some } i \neq j\}.$$

At fixed  $\alpha$ , define

$$i \sim_\alpha j \iff \lambda_i(\alpha) = \lambda_j(\alpha).$$

The partition  $P/\sim_\alpha$  is the collision quotient. Write

$$N_P(\alpha) = |P/\sim_\alpha|, \quad \Delta_P(\alpha) = N - N_P(\alpha).$$

**Definition 2.3** (Pair multiplicity and quotient drop).

Let  $C_1, \dots, C_r$  be the non-singleton classes of  $P/\sim_\alpha$ . Define

$$m_{\text{pair}}(\alpha) = \sum_{a=1}^r \binom{|C_a|}{2}, \quad \Delta_P(\alpha) = \sum_{a=1}^r (|C_a| - 1).$$

These two quantities are distinct. For example, a triple collision contributes three pair equalities but only two units of layer-count drop.

**Definition 2.4** (Branch order).

For  $\alpha \notin \mathcal{C}(P)$ , the values  $\lambda_i(\alpha)$  are distinct. Their descending order defines the branch order at  $\alpha$ . A chamber is a connected component of  $\mathbb{R} \setminus \mathcal{C}(P)$ .

The quotient partition is discrete on every chamber. The branch order is the additional datum that distinguishes neighboring chambers.

### 3 General Collision–Quotient Theorem

**Theorem 3.1** (Finite collision arrangement).

Let  $P = \{(q_i, h_i)\}_{i=1}^N \subset \mathbb{R}^2$  be a finite set of distinct points.

1. If the branches  $i$  and  $j$  have different slopes, they collide at the unique parameter

$$\alpha_{ij} = \frac{h_j - h_i}{(q_i - h_i) - (q_j - h_j)}.$$

2. If the slopes agree, the distinct branches are parallel and never collide.
3. The collision set  $\mathcal{C}(P)$  is finite.
4. On each chamber of  $\mathbb{R} \setminus \mathcal{C}(P)$ , the quotient is discrete and the strict branch order is constant.
5. At a critical parameter, the non-singleton collision classes are exactly the connected components of the graph whose edges are the pairs satisfying  $\alpha_{ij} = \alpha$ .

**Proof.** Equality of two branches gives

$$h_i + \alpha(q_i - h_i) = h_j + \alpha(q_j - h_j).$$

If the coefficient of  $\alpha$  is nonzero, solving for  $\alpha$  yields the displayed formula and guarantees uniqueness. If the coefficient is zero, the slopes agree. The distinctness of the points then implies  $h_i \neq h_j$ , so the parallel affine functions never meet. Because there are only  $\binom{N}{2}$  pairs, the collision set is finite.

Outside  $\mathcal{C}(P)$ , every difference  $\lambda_i - \lambda_j$  is nonzero. Its sign is continuous and therefore constant on each connected chamber. Consequently, the strict branch order remains constant within each chamber. At a critical parameter, equality is transitive; thus, the equality classes correspond exactly to the connected components of the equal-label graph.

**Theorem 3.2** (Spectral layers as collision classes).

Let  $Q, H$  be exactly commuting Hermitian operators on a finite-dimensional complex Hilbert space. Let

$$V = \bigoplus_{i=1}^N E_i, \quad Q|_{E_i} = q_i I, \quad H|_{E_i} = h_i I,$$

where equal joint eigenvalue pairs have already been combined. Then the eigenspaces of

$$A(\alpha) = \alpha Q + (1 - \alpha)H$$

are precisely

$$\bigoplus_{i \in C} E_i, \quad C \in P / \sim_\alpha.$$

**Proof.** On  $E_i$ , the operator  $A(\alpha)$  acts by the scalar  $L_\alpha(q_i, h_i)$ . Two joint eigenspaces belong to the same eigenspace of  $A(\alpha)$  exactly when their scalar values agree.

**Boundary.** This theorem assumes exact commutation and exact joint spectral data. A small numerical commutator residual does not establish its hypotheses.

## 4 Rational Arrangements

**Proposition 4.1** (Rational collision parameters).

*If  $P \subset \mathbb{Q}^2$  is finite, every finite collision parameter belongs to  $\mathbb{Q}$ .*

**Proof.** *The collision formula uses subtraction and division by a nonzero rational number.*

*We do not claim a denominator bound that depends solely on a common coordinate denominator. Such a bound would also require a declared input height or coordinate range. For the explicit arrangement studied below, small denominators are established through a direct exact census rather than by invoking a general height theorem. The comparison with Farey-type small-denominator organization is descriptive only [7].*

**Proposition 4.2** (Rectangular control family).

*For  $a, b > 0$ , let*

$$P_{\text{rect}}(a, b) = \{p_{00} = (0, 0), p_{10} = (a, 0), p_{01} = (0, b), p_{11} = (a, b)\}.$$

*The unique collision parameter in the open interval  $(0, 1)$  is*

$$\alpha_* = \frac{b}{a + b}.$$

*At  $\alpha_*$ , the only non-singleton quotient class is  $\{p_{10}, p_{01}\}$ , so  $N_{P_{\text{rect}}}(\alpha_*) = 3$  and  $\Delta_{P_{\text{rect}}}(\alpha_*) = 1$ . At every other parameter in  $(0, 1)$ , the four branch values are distinct. If  $a, b \in \mathbb{Q}_{>0}$ , then  $\alpha_* \in \mathbb{Q}$ .*

**Proof.** *The four branches are*

$$0, \quad \alpha a, \quad (1 - \alpha)b, \quad \alpha a + (1 - \alpha)b.$$

*For  $0 < \alpha < 1$ , the first is strictly smaller and the last strictly larger than both middle branches. The middle branches agree exactly when  $\alpha a = (1 - \alpha)b$ , which gives the displayed value. Rationality follows immediately when  $a$  and  $b$  are rational.*

*This family shows that an interior collision quotient is not peculiar to the nine-point census below. It also separates a general adjustable collision parameter from the special multiplicity and maximal-drop properties of  $P_9$ .*

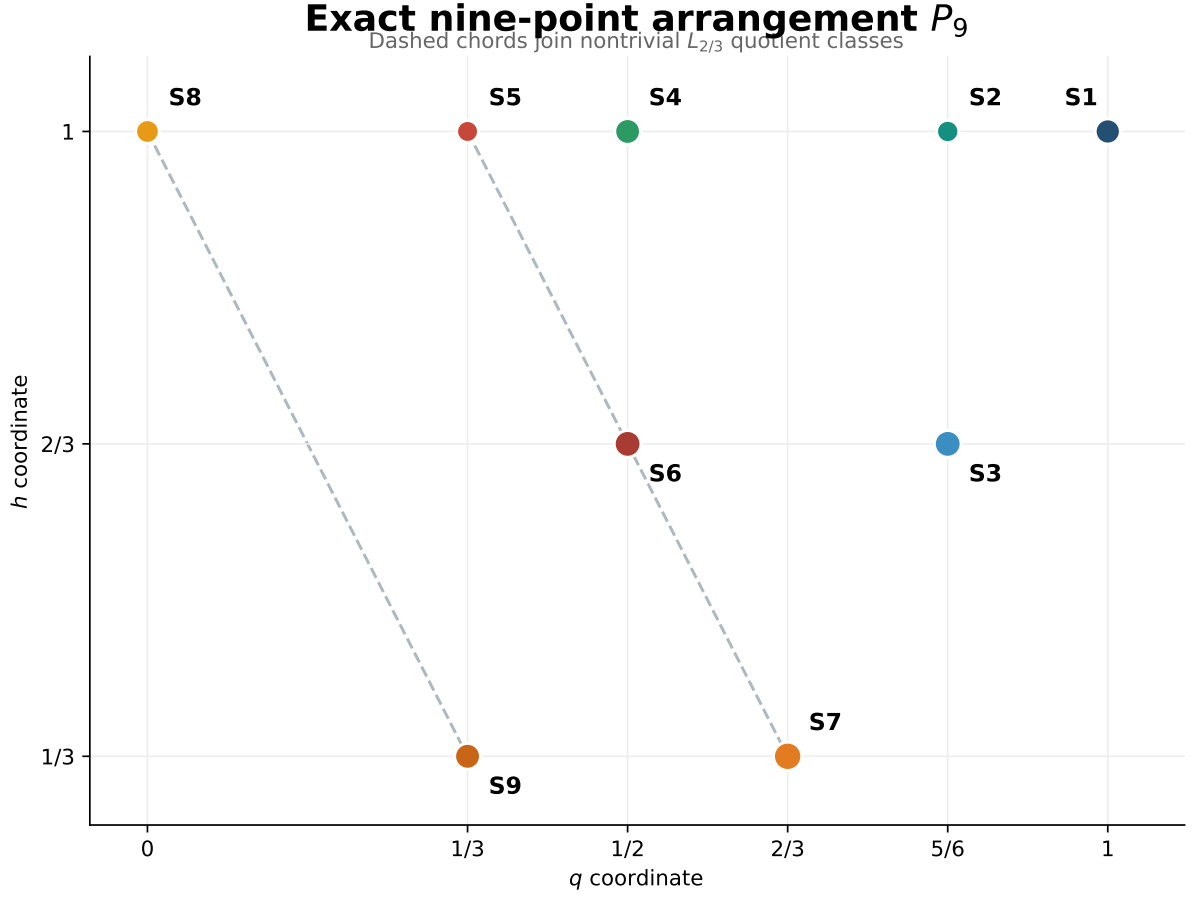
## 5 Exact Nine-Point Census

Define the following weighted rational arrangement independently of any operator realization:

$$P_9 = \{(S_i, d_i, q_i, h_i) : 1 \leq i \leq 9\}.$$

| Label | Weight $d_i$ | $q_i$ | $h_i$ | $q_i - h_i$ | $L_{2/3}(q_i, h_i)$ |
|-------|--------------|-------|-------|-------------|---------------------|
| S1    | 20           | 1     | 1     | 0           | 1                   |
| S2    | 2            | 5/6   | 1     | -1/6        | 8/9                 |
| S3    | 39           | 5/6   | 2/3   | 1/6         | 7/9                 |
| S4    | 26           | 1/2   | 1     | -1/2        | 2/3                 |
| S5    | 1            | 1/3   | 1     | -2/3        | 5/9                 |
| S6    | 39           | 1/2   | 2/3   | -1/6        | 5/9                 |
| S7    | 66           | 2/3   | 1/3   | 1/3         | 5/9                 |
| S8    | 8            | 0     | 1     | -1          | 1/3                 |
| S9    | 27           | 1/3   | 1/3   | 0           | 1/3                 |

The weights sum to 228. In this section, they are attached integer weights; their interpretation as operator ranks belongs to the computational registration and conditional corollary below.



**Figure 1.** The independent rational arrangement  $P_9$  in the  $(q, h)$  plane. Sector labels identify the nine declared points. Dashed chords mark the nontrivial  $L_{2/3}$  quotient classes of this exact finite arrangement; the figure does not certify an exact Rubik joint spectrum.

**Theorem 5.1** (Exact nine-point census).

*For the displayed rational arrangement:*

1. the 36 unordered pairs split into 2 parallel, 10 interior, 15 endpoint, and 9 exterior pairs;
2. the interior collision set on  $[0, 1]$  is

$$\{2/7, 2/5, 1/2, 2/3, 4/5\};$$

3. the critical quotient data are:

| $\alpha$ | Classes                      | $N_P(\alpha)$ | $\Delta_P(\alpha)$ | $m_{\text{pair}}(\alpha)$ |
|----------|------------------------------|---------------|--------------------|---------------------------|
| 0        | S1=S2=S4=S5=S8; S3=S6; S7=S9 | 3             | 6                  | 12                        |
| 2/7      | S3=S8                        | 8             | 1                  | 1                         |
| 2/5      | S3=S5; S6=S8                 | 7             | 2                  | 2                         |
| 1/2      | S3=S4; S7=S8                 | 7             | 2                  | 2                         |
| 2/3      | S5=S6=S7; S8=S9              | 6             | 3                  | 4                         |
| 4/5      | S4=S7                        | 8             | 1                  | 1                         |
| 1        | S2=S3; S4=S6; S5=S9          | 6             | 3                  | 3                         |

4.  $\alpha = 2/3$  is the unique **interior** parameter maximizing the layer-count drop  $\Delta_P$ .

**Proof.** Appendix A lists all  $\binom{9}{2} = 36$  pairs, their exact  $\alpha_{ij}$  values, and their classifications. The table contains 2 parallel, 10 interior, 15 endpoint, and 9 exterior pairs. Grouping equal interior values yields  $\{2/7, 2/5, 1/2, 2/3, 4/5\}$ . Direct substitution of the nine displayed points at the seven parameters in the critical table produces the quotient classes, sizes, drops, and pair multiplicities. The interior drops are 1, 2, 2, 3, 1, so the interior maximum is uniquely attained at  $2/3$ . Every step uses only the displayed rational coordinates and the collision formula.

**Corollary 5.2** (Exact weighted quotient at alpha two-thirds).

*At  $\alpha = 2/3$ , the weighted quotient of  $P_9$  is*

| Value | Members    | Total weight |
|-------|------------|--------------|
| 1     | S1         | 20           |
| 8/9   | S2         | 2            |
| 7/9   | S3         | 39           |
| 2/3   | S4         | 26           |
| 5/9   | S5, S6, S7 | 106          |
| 1/3   | S8, S9     | 35           |

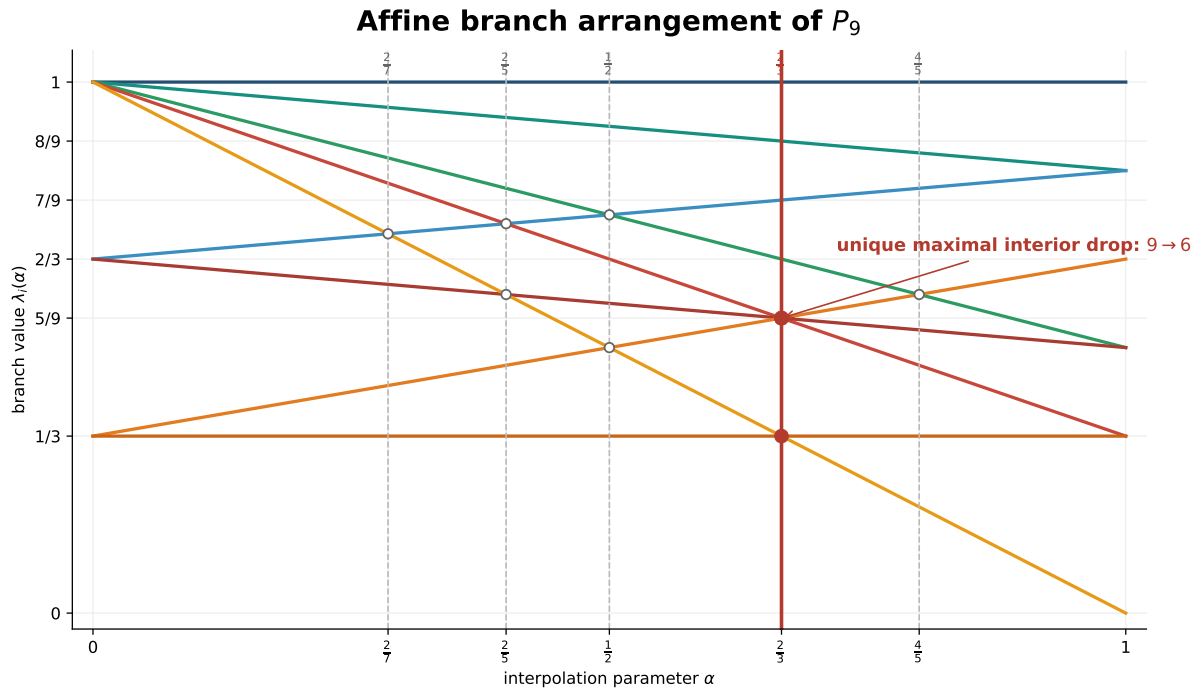
This is an exact statement about the displayed weighted point set.

## 5.1 Chamber order on the unit interval

Every open chamber has nine distinct values. The descending branch orders are:

| Chamber    | Descending order                   |
|------------|------------------------------------|
| (0, 2/7)   | S1, S2, S4, S5, S8, S3, S6, S7, S9 |
| (2/7, 2/5) | S1, S2, S4, S5, S3, S8, S6, S7, S9 |
| (2/5, 1/2) | S1, S2, S4, S3, S5, S6, S8, S7, S9 |
| (1/2, 2/3) | S1, S2, S3, S4, S5, S6, S7, S8, S9 |
| (2/3, 4/5) | S1, S2, S3, S4, S7, S6, S5, S9, S8 |
| (4/5, 1)   | S1, S2, S3, S7, S4, S6, S5, S9, S8 |

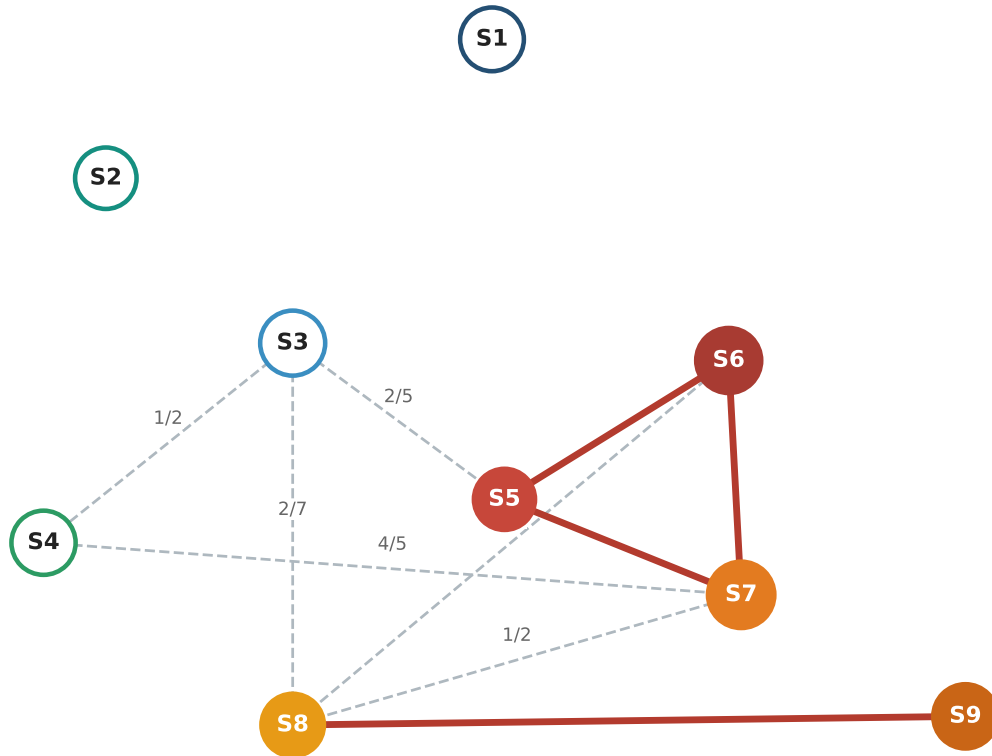
Together with the critical table, this is the full interpolation phase diagram of  $P_9$  on  $[0, 1]$ .



**Figure 2.** Affine branches of the independent arrangement  $P_9$ . The five interior critical parameters and the unique interior maximum of the layer-count drop are exact finite-point statements. Their identification with an actual Rubik joint spectrum is conditional on the registration hypotheses below.

## Interior collision graph of $P_9$

Red edges occur at  $\alpha = 2/3$ ; gray edges mark the other interior collisions



**Figure 3.** Interior collision graph of  $P_9$ . Red edges have label  $2/3$  and form the exact triple class  $S5=S6=S7$  together with  $S8=S9$ . Gray dashed edges mark the other exact interior collisions of the displayed arrangement.

## 6 Computational Rubik Registration

In the declared 228-dimensional complex128 cubie realization, define

$$Q_{\text{num}} = QT_{\text{all}}, \quad H_{\text{num}} = HT_{\text{all}}.$$

The generator counts give the definitional identity

$$A_{18} = \frac{12Q_{\text{num}} + 6H_{\text{num}}}{18} = \frac{2}{3}Q_{\text{num}} + \frac{1}{3}H_{\text{num}}.$$

This identity does not imply that  $Q_{\text{num}}$  and  $H_{\text{num}}$  commute exactly or that their numerical clusters have exact rational eigenvalues.

### 6.1 Registration algorithm

For each declared clustering tolerance  $\tau$ , the registration is the following deterministic numerical procedure. First diagonalize  $Q_{\text{num}}$  and partition its ordered numerical eigenvalues into clusters by the absolute rule

$$|\hat{q}_j - \hat{q}_i| < \tau,$$

where  $\hat{q}_i$  is the first eigenvalue in the cluster. Within each resulting  $Q_{\text{num}}$  eigenspace, diagonalize the restricted  $H_{\text{num}}$  and apply the same absolute rule. The resulting orthonormal basis  $B_i$  defines

$$P_i = B_i B_i^*, \quad \hat{q}_i = \frac{\text{tr}(B_i^* Q_{\text{num}} B_i)}{\dim B_i}, \quad \hat{h}_i = \frac{\text{tr}(B_i^* H_{\text{num}} B_i)}{\dim B_i}.$$

Thus the representative is a mean Rayleigh scalar and the rank is the number of orthonormal basis columns. The nine raw coordinate pairs are then matched one-to-one to the predeclared set  $P_9$  by a Hungarian assignment with cost

$$d_{\infty}((q, h), (q', h')) = \max\{|q - q'|, |h - h'|\}.$$

There is no random step, denominator search, or rational reconstruction: the procedure measures only agreement with the independently declared table.

#### **Computational Certificate 6.1 (Registered QT/HT clusters).**

In the declared complex128 construction, the registration algorithm above produces nine clusters with coordinate-matched ranks

$$(20, 2, 39, 26, 1, 39, 66, 8, 27).$$

Their raw scalar coordinates match the rational coordinates of  $P_9$ . The maximum raw-to-table discrepancy across the QT, HT, and  $A_{18}$  coordinates is

$$9.992 \times 10^{-16}.$$

The principal numerical diagnostics are:

| Diagnostic                                      | Frobenius residual      |
|-------------------------------------------------|-------------------------|
| $Q_{\text{num}} - Q_{\text{num}}^*$             | $2.73 \times 10^{-16}$  |
| $H_{\text{num}} - H_{\text{num}}^*$             | 0                       |
| $[Q_{\text{num}}, Q_{\text{num}}^*]$            | $3.17 \times 10^{-18}$  |
| $[H_{\text{num}}, H_{\text{num}}^*]$            | 0                       |
| $[Q_{\text{num}}, H_{\text{num}}]$              | $3.77 \times 10^{-16}$  |
| $A_{18} - (2Q_{\text{num}} + H_{\text{num}})/3$ | $2.94 \times 10^{-17}$  |
| $\max_i \ P_i^2 - P_i\ _F$                      | $1.087 \times 10^{-14}$ |
| $\max_i \ P_i - P_i^*\ _F$                      | $6.314 \times 10^{-16}$ |
| $\max_{i < j} \ P_i P_j\ _F$                    | $4.782 \times 10^{-15}$ |
| $\ \sum_i P_i - I\ _F$                          | $2.229 \times 10^{-14}$ |
| maximum rank–trace discrepancy                  | $1.421 \times 10^{-14}$ |
| maximum joint-eigen residual                    | $8.661 \times 10^{-15}$ |

For  $\tau \in \{10^{-6}, 10^{-8}, 10^{-10}, 10^{-12}\}$ , the audit returns nine clusters with the same coordinate-matched label–rank pairs, zero coordinate and projector drift relative to  $\tau = 10^{-10}$  at printed precision, and minimum normalized trace overlap

$$\min_i \frac{\text{tr}(P_i(\tau)P_i(10^{-10}))}{\text{rank } P_i} = 0.9999999999999983.$$

The predeclared arrangement has the exact separation

$$\min_{i \neq j} \|p_i - p_j\|_\infty = \frac{1}{6}.$$

The complete tolerance certificate appears in Appendix B.

**Certificate boundary.** This finite computation certifies numerical stability and proximity to the declared rational table for the declared complex128 realization, clustering rule, tolerances, and implementation. It does not establish any exact hypothesis used by Corollary 7.1: in particular, it does not prove exact commutation, exact rational joint spectral data, or exact primitive idempotents.

Within this numerical layer, S1–S9 are called the **registered joint clusters**. The phrase “primitive idempotent of  $\mathbb{C}[Q, H]$ ” is reserved for the exact conditional setting.

## 7 Conditional Rubik Interpretation

The promotion from Computational Certificate 6.1 to an exact Rubik statement is governed by the following three independently checkable assumptions.

**Assumption R1 (exact operator admissibility).** The exact operators  $Q = QT_{\text{all}}$  and  $H = HT_{\text{all}}$  are Hermitian and commute:

$$Q = Q^*, \quad H = H^*, \quad [Q, H] = 0.$$

**Assumption R2 (exact labelled joint registration).** There is an orthogonal decomposition

$$V = \bigoplus_{i=1}^9 E_i$$

such that  $\dim E_i = d_i$ ,  $Q|_{E_i} = q_i I$ , and  $H|_{E_i} = h_i I$ , with the labelled triples  $(d_i, q_i, h_i)$  exactly equal to the displayed  $P_9$  table.

**Assumption R3 (exact averaging identity).** The exact canonical average satisfies

$$A_{18} = \frac{2Q + H}{3}.$$

When  $Q$  and  $H$  are defined as the equal-weight averages of the twelve quarter turns and six half turns, respectively, R3 is the corresponding exact finite-sum identity. Its numerical residual in Certificate 6.1 is an implementation check, not a substitute for that exact definition.

The checklist is conjunctive: the conclusion below is available only when R1, R2, and R3 all hold. Certificate 6.1 supplies numerical evidence relevant to R1 and R2 but discharges neither assumption.

**Corollary 7.1** (Conditional Rubik collision quotient).

*Under Assumptions R1–R3, the eigenspace partition of  $A_{18}$  is the direct-sum partition induced by the  $L_{2/3}$  collision quotient of  $P_9$ . In particular, the exact eigenvalues and eigenspace dimensions are*

$$(1, 20), (8/9, 2), (7/9, 39), (2/3, 26), (5/9, 106), (1/3, 35).$$

**Proof.** Apply Theorem 3.2 and Corollary 5.2.

*Under these hypotheses, the nine exact joint projectors are the primitive idempotents of the represented algebra  $\mathbb{C}[Q, H]$ . This primitivity is relative to the declared algebra only. It does not assert maximality among larger commuting extensions in  $\text{End}(V)$ .*

*The corollary explains the relationship between the registered six-layer census and the finer nine-cluster table. However, it does not prove the exact registration hypotheses, nor does it provide an independent arithmetic proof of canonical Rubik rationality. The corresponding six-layer census has been recorded elsewhere as a finite computation with mixed analytic and numerical input [3].*

## 8 Negative Control

Collision adjacency depends only on equality of projected scalar values. Transport and composition require an additional labelled operative family and its projected blocks. The distinction already has an exact negative control.

**Proposition 8.1** (Diagonal negative control).

*Let  $P = \{(q_i, h_i)\}_{i=1}^N$  be any finite arrangement with a nontrivial collision  $i \sim_\alpha j$ . On  $V = \mathbb{C}^N$ , let  $Q$  and  $H$  be diagonal with diagonal entries  $q_i$  and  $h_i$ , and let every operator in a declared operative family  $\mathcal{Y}$  be diagonal in the same basis. Then the collision edge  $i$ – $j$  is present, while every off-diagonal direct-support block and every projected composition between distinct sectors is zero.*

**Proof.** For the coordinate projectors  $P_i$ , diagonality gives  $P_i Y P_j = 0$  whenever  $i \neq j$  and  $Y \in \mathcal{Y}$ . Every projected product with distinct endpoint sectors contains such a diagonal off-sector block and therefore vanishes. The equality  $L_\alpha(q_i, h_i) = L_\alpha(q_j, h_j)$  is unchanged because it depends only on  $Q$  and  $H$ .

*Thus collision adjacency alone implies neither transport adjacency nor a nonzero routed composition. The Rubik registration gives a less degenerate finite comparison. For the explicit arrangement, the 2/3 collision graph contains the complete triangle*

$$S5-S6, \quad S5-S7, \quad S6-S7.$$

## 8.1 Triangle versus chain

### Computational Observation 8.2 (Rubik triangle versus support chain).

In the declared Rubik numerical registration, direct generator-support blocks are nonzero for S5–S6 and S6–S7 and below the declared threshold for S5–S7. Thus the registered direct-support graph on these three clusters is the chain

$$S5-S6-S7,$$

not the collision triangle.

**No-promotion boundary.** The exact negative control and the finite Rubik observation have different roles. Proposition 8.1 refutes any general implication from collision adjacency to transport or composition. Observation 8.2 records a nontrivial numerical separation in the declared Rubik realization, but it does not establish a general transport theorem. Collision geometry records equality of affine scalar values, whereas direct support records nonzero projected generator blocks. Even a direct-support chain does not by itself certify a nonzero projected composition; that step requires a separate image–kernel audit [4]. If sectors are merged, graph incidence and operator products must be explicitly recomputed at the new resolution rather than automatically pushed forward through the collision quotient.

## 9 Claim Status and Boundary

The table uses the four claim levels. Exact finite theorems and conditional corollaries remain in the Theorem level under their displayed hypotheses.

| Claim                                                                  | Status                    |
|------------------------------------------------------------------------|---------------------------|
| Collision formula and finite critical set                              | Theorem                   |
| Chamberwise constancy of branch order                                  | Theorem                   |
| Spectral layers equal collision classes for commuting Hermitian $Q, H$ | Theorem                   |
| Rational coordinates imply rational finite collision parameters        | Theorem                   |
| Rectangular family has one adjustable interior collision               | Theorem                   |
| Complete census of the displayed $P_9$                                 | Theorem                   |
| $2/3$ is the unique interior maximal-drop point of $P_9$               | Theorem                   |
| Numerical QT/HT clusters match $P_9$ to declared residuals             | Computational Certificate |
| Exact Rubik eigenspaces are the $2/3$ quotient under R1–R3             | Theorem                   |
| Collision adjacency does not generally imply transport or composition  | Theorem                   |
| Collision triangle differs from direct-support chain                   | Computational Observation |
| QH algebra is canonical or maximal among commuting refinements         | Research Program          |
| Moving arrangements and spectral-wall inclusions                       | Research Program          |
| Spectral coarse-graining preserves information flow                    | Research Program          |

The principal failure boundary is the promotion step:

$$\text{numerical cluster registration} \not\Rightarrow \text{exact joint spectral theorem.}$$

Exact arithmetic on the predeclared table validates that table. It does not retroactively prove that the numerical operator data are exact rational spectral data. Likewise, a PASS result for Certificate 6.1 does not mark R1 or R2 as proved; the three-item checklist in Section 7 remains external to that certificate.

## 10 Related Work and Novelty Boundary

Simultaneous diagonalization and joint eigenspaces are standard matrix analysis [8]. This fixed commuting setting differs from general Hermitian pencils and analytic perturbation theory: eigenvalue branches need not be affine, and noncommuting perturbations can produce avoided crossings. Classical perturbation theory supplies that broader boundary; see [10] and [13].

Pair equalities

$$L_\alpha(p_i - p_j) = 0$$

are one-dimensional sections of a hyperplane discriminant arrangement. The language of hyperplane arrangements and discriminants provides the natural geometric comparison class [11]. This paper does not claim a new general eigenvalue-perturbation theorem. Its contribution is the explicit organization of a fixed joint-spectrum projection as a collision quotient, the rectangular control family, the exact census of  $P_9$ , and the claim-status-separated Rubik registration.

Association schemes provide a second comparison class: their Bose–Mesner algebras are finite-dimensional commutative semisimple algebras organized by primitive idempotents, and quotient schemes arise from additional equitable partition structure [1, 5, 6]. No association-scheme or coherent-configuration structure is claimed for the registered QT/HT pair. Godsil–Martin quotient theory provides a comparison analogy, not a proof of Theorem 3.2.

The Rubik operators are represented group-algebra averages in the standard finite-group and cubie setting [12, 9]. Related studies record the block-spectral census, direct transport on the registered QT/HT clusters, and the separation between support-graph paths and projected matrix composition [3, 2, 4]. These results are not used in the general finite-point proofs.

## 11 Research Program

The following directions are not established in this article.

### 11.1 Moving arrangements

Generator deletion or reweighting may replace the fixed arrangement  $P$  by a moving family  $P(w)$ . Before collision quotients can be assigned, a deformation theory must pass the gates

$$\begin{array}{c} \text{commutativity} \longrightarrow \text{normality} \longrightarrow \text{spectral chart} \\ \downarrow \\ \text{orthogonal joint projectors} \longrightarrow \text{candidate layer and field walls} \end{array}$$

This sequence remains a proposed programmatic outline. The existence of such charts, constancy of rank, projector continuation, and wall inclusions are not established here. Commutativity alone does not supply normality or orthogonal joint projectors.

### 11.2 Candidate wall hierarchy

Symbols such as  $\Sigma_{\text{comm}}$ ,  $\Sigma_{\text{spec}}$ ,  $\Sigma_L$ , and  $\Sigma_{\text{field}}$  belong to a deformation program. No subset chain among them is asserted here. Establishing a valid hierarchy would require definitions on a common admissible weight space and proofs of each domain inclusion.

### 11.3 Maximal commuting refinements

The numerical QH clusters are minimal at the declared clustering tolerance. Under the exact hypotheses of Corollary 7.1, the exact joint projectors would be primitive relative to  $\mathbb{C}[Q, H]$ . Whether this algebra is canonical or maximal in a symmetry-constrained class is open.

### 11.4 Higher-dimensional arrangements

For  $P \subset \mathbb{R}^d$  and a weight vector  $w$ , pair collisions lie on the hyperplanes  $w \cdot (p_i - p_j) = 0$ . Developing the corresponding chamber and quotient theory beyond the one-dimensional interpolation studied here is a natural extension.

### 11.5 Spectral coarse-graining and information flow

The triangle-versus-chain observation shows that a spectral quotient can hide resolved direct-support data. A future spectral coarse-graining or “spectral RG” theory would need explicit criteria for which support, word-composition, or Lie-accessibility data survive quotienting. While the Rubik triangle-versus-chain comparison motivates this question, it does not establish a general information-flow theorem.

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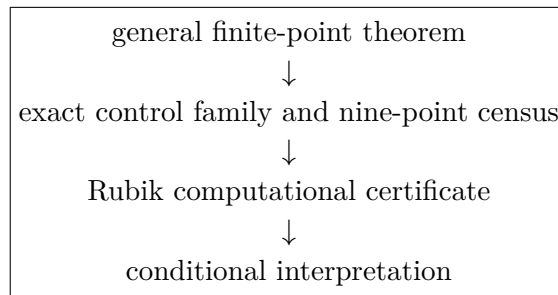
## 12 Conclusion

For a fixed finite arrangement, collision quotients are exact finite objects. The critical set is finite, the branch order is chamberwise constant, and a commuting-Hermitian realization identifies quotient classes with spectral layers. The rectangular family supplies an adjustable one-collision control. The explicit rational set  $P_9$  admits a complete exact census on  $[0, 1]$ , with  $2/3$  the unique interior parameter of maximal layer-count drop.

The Rubik interpretation has a different status. The declared numerical operators produce nine stable clusters extremely close to  $P_9$ , which is a Computational Certificate. Numerical residuals and table agreement do not prove R1 or R2. Only under the conjunctive assumptions R1–R3 does Corollary 7.1 identify the exact  $A_{18}$  eigenspace partition with the  $L_{2/3}$  collision quotient of the nine-point arrangement.

Finally, collision data do not carry transport or composition semantics. The diagonal negative control proves the general nonimplication, while the Rubik triangle-versus-chain comparison remains a separate Computational Observation.

The resulting architecture is therefore:



Moving arrangements, normal spectral charts, maximal refinements, and information-flow preservation begin only after this fixed-arrangement theory.

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## Appendix A Complete Pair Certificate

This appendix gives a human-checkable certificate for Theorem 5.1. For each unordered pair, the entry is obtained by substituting the displayed rational coordinates into Theorem 3.1. A dash means equal slopes and hence no finite collision. The four tables contain all 36 pairs exactly once.

### A.1 Parallel pairs

| Pair  | $\alpha_{ij}$ | Classification |
|-------|---------------|----------------|
| S1–S9 | –             | parallel       |
| S2–S6 | –             | parallel       |

### A.2 Interior pairs

| Pair  | $\alpha_{ij}$ | Classification |
|-------|---------------|----------------|
| S3–S4 | 1/2           | interior       |
| S3–S5 | 2/5           | interior       |
| S3–S8 | 2/7           | interior       |
| S4–S7 | 4/5           | interior       |
| S5–S6 | 2/3           | interior       |
| S5–S7 | 2/3           | interior       |
| S6–S7 | 2/3           | interior       |
| S6–S8 | 2/5           | interior       |
| S7–S8 | 1/2           | interior       |
| S8–S9 | 2/3           | interior       |

### A.3 Endpoint pairs

| Pair  | $\alpha_{ij}$ | Classification |
|-------|---------------|----------------|
| S1–S2 | 0             | endpoint       |
| S1–S4 | 0             | endpoint       |
| S1–S5 | 0             | endpoint       |
| S1–S8 | 0             | endpoint       |
| S2–S3 | 1             | endpoint       |
| S2–S4 | 0             | endpoint       |
| S2–S5 | 0             | endpoint       |
| S2–S8 | 0             | endpoint       |
| S3–S6 | 0             | endpoint       |
| S4–S5 | 0             | endpoint       |
| S4–S6 | 1             | endpoint       |
| S4–S8 | 0             | endpoint       |
| S5–S8 | 0             | endpoint       |
| S5–S9 | 1             | endpoint       |
| S7–S9 | 0             | endpoint       |

## A.4 Exterior pairs

| Pair  | $\alpha_{ij}$ | Classification |
|-------|---------------|----------------|
| S1–S3 | 2             | exterior       |
| S1–S6 | −2            | exterior       |
| S1–S7 | 2             | exterior       |
| S2–S7 | 4/3           | exterior       |
| S2–S9 | 4             | exterior       |
| S3–S7 | 2             | exterior       |
| S3–S9 | −2            | exterior       |
| S4–S9 | 4/3           | exterior       |
| S6–S9 | 2             | exterior       |

The counts are therefore  $2 + 10 + 15 + 9 = 36$ . Grouping the ten interior rows by their exact parameter gives multiplicities 1, 2, 2, 4, 1 at  $2/7, 2/5, 1/2, 2/3, 4/5$ , respectively. The quotient-class table in Theorem 5.1 then follows by transitive closure at each common parameter.

## Appendix B Registration Stability Certificate

The baseline registration uses  $\tau = 10^{-10}$ . Across all four tested tolerances, coordinate matching to  $P_9$  returns the same label–rank list

$$(S1, 20), (S2, 2), (S3, 39), (S4, 26), (S5, 1), (S6, 39), (S7, 66), (S8, 8), (S9, 27).$$

The projector comparison uses the labels assigned by the global  $d_\infty$  matching, not a sorted rank census.

| $\tau$     | Clusters | Label–rank stable | Max $d_\infty$ match to $P_9$ | Max coordinate drift |
|------------|----------|-------------------|-------------------------------|----------------------|
| $10^{-6}$  | 9        | yes               | $9.992 \times 10^{-16}$       | 0                    |
| $10^{-8}$  | 9        | yes               | $9.992 \times 10^{-16}$       | 0                    |
| $10^{-10}$ | 9        | yes               | $9.992 \times 10^{-16}$       | 0                    |
| $10^{-12}$ | 9        | yes               | $9.992 \times 10^{-16}$       | 0                    |

| $\tau$     | Max projector drift in Frobenius norm | Min normalized overlap |
|------------|---------------------------------------|------------------------|
| $10^{-6}$  | 0                                     | 0.9999999999999983     |
| $10^{-8}$  | 0                                     | 0.9999999999999983     |
| $10^{-10}$ | 0                                     | 0.9999999999999983     |
| $10^{-12}$ | 0                                     | 0.9999999999999983     |

Zeros denote values returned in floating-point complex128 arithmetic, not exact symbolic identities. The overlap is

$$\frac{\text{tr}(P_i(\tau)P_i(10^{-10}))}{\text{rank } P_i}.$$

The implementation records the clustering rule, matching metric, dtype, package versions, source hashes, Git state, and runtime in the observation artifact. These data certify reproducibility of the stated numerical audit; they do not promote numerical projectors or coordinates to exact objects.

## Appendix C Computational Artifacts

The following repository artifacts support the exact finite census and the computational Rubik case study. The default directory is `experiments/paper4/`; paths are relative to that directory.

| Artifact | Role                                                                 | Short path                                                                   |
|----------|----------------------------------------------------------------------|------------------------------------------------------------------------------|
| C1       | exact <code>Fraction</code> census of the declared $P_9$ arrangement | <code>validation/rubik_collision_quotient.py</code>                          |
| C2       | numerical Rubik registration and residual audit                      | <code>validation/rubik_joint_spectrum_registration.py</code>                 |
| C3       | collision-triangle/direct-support-chain comparison                   | <code>validation/v59_collision_vs_transport.py</code>                        |
| C4       | source-addressed v2.1 registration observation                       | <code>results/rubik_joint_spectrum_registration_v2_1_observation.json</code> |

C2 reports operator Hermiticity and normality, QT/HT commutation,  $A_{18}$  reconstruction, projector identities, ranks, joint-eigen residuals, raw-to- $P_9$  discrepancies,  $\ell_\infty$  separation, and matched-projector stability. From the repository root, run an executable artifact as `python experiments/paper4/<short path>`; append `--check-result` to C2 to check C4 against its declared source hashes.

C4 is a source-addressed run record, not an independent proof. Reproducing the certificate requires a full run of C2.

All listed artifacts are available in the [RIME repository](#).

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